GA4 GOOGLE MERCHANDISE STORE

Funnel and Retention Analysis

Business findings and decision analysis

Observation period: 1 November 2020 - 31 January 2021

Prepared 12 October 2026 | Revised 13 October 2026

For ecommerce, product, marketing and analytics decision-makers

Historical, obfuscated public sample. Findings describe the captured data; proposed actions require validated production evidence.

Power BI dashboard + Streamlit evidence-based chat

Problem statement

Using the GA4 Google Merchandise Store sample dataset, identify where users drop out of the purchase funnel, how conversion varies across months and devices, and how often users return. Assess data quality so decision-makers can distinguish reliable findings from measurement limitations and prioritize improvements.

Business questions

1. Where is the largest measured funnel loss, and which stage should be investigated first?
2. How much of January's purchase-session decline reflects traffic volume versus conversion?
3. Do device differences justify a device-specific intervention?
4. What do mature cohorts reveal about return activity and repeat purchasing?
5. Which acquisition, transaction and item-quality issues limit commercial decisions?
6. What test could assess an improvement, with suitable outcomes and guardrails?

Method and scope

The analysis reconciles preserved GA4 aggregates, compares ordered session-based funnel stages, separates traffic and conversion changes, compares monthly device segments, and uses fully observed calendar-week cohorts. Acquisition is first-user scoped. Data-quality checks and metric denominators are reported beside the findings.

No charts are included in this edition. Numerical evidence is presented in tables. The source register and current dashboard reconciliation appear at the end.

Prioritized recommendations

Owners are proposed roles. Priorities indicate sequence, not promised dates or approved commitments.

Priority	Action	Proposed owner	Success metric	Dependencies
P1	Validate measurement	Analytics engineering + ecommerce product	Reconciled orders; agreed event coverage; resolved identity exceptions	Known desktop/mobile test journeys; trusted order system
P1	Diagnose January decline	Product analyst + ecommerce product	Daily stage rates and coverage reviewed; evidence-backed cause hypotheses	Release/tag history; stock, assortment and promotion records
P2	Test one cart-action change	Product + design + experimentation	Successful ordered cart per exposed user; mature purchase and error guardrails	Validated production user baseline; stable assignment; predeclared design
P3	Evaluate acquisition	Marketing analytics	Incremental accepted purchases and commercial return	Compatible spend/order attribution; source-quality fixes
P3	Test repeat-purchase engagement	CRM + product analytics	Incremental mature accepted repeat purchases; unsubscribe guardrail	Consent/contactability; product-cycle evidence; mature user holdout
P3	Revisit item/category decisions	Analytics engineering + merchandising	Stable item/variant linkage and reconciled outcomes	Validated item payloads and category mapping

Decision gate

Validate measurement before launching a product-page test. Withhold acquisition budget and category investment changes until attribution, commercial outcomes and item linkage are dependable.

Executive Assessment

The strongest practical recommendation is to validate the measurement of the shopping journey and then test an improvement to the product-view-to-cart step. This step has the largest observed loss of sessions. A second priority is to investigate the January deterioration, which cannot be explained by lower traffic alone. The data does not support reallocating acquisition budgets, concluding that mobile is broken, or making product-category investment decisions.

The report describes a historical, obfuscated sample of the Google Merchandise Store, not current business performance. Google explicitly warns that obfuscation limits internal consistency. Consequently, the findings are useful for identifying questions, designing experiments and explaining the dashboard; they are not a forecast of the actual store's sales or proof of a specific operational failure. [R1]

Five findings should drive the discussion:

1. Product views rarely progress to an ordered cart event. Of 77,020 product-view sessions, 14,380 reach cart, a conversion of 18.67%. The remaining 62,640 sessions do not meet that next step in the same session. This is the largest observed opportunity by volume.

2. January purchase-session volume fell 47.28% from December, although traffic fell only 11.23%. Purchase-session conversion fell from 1.586% to 0.942%. Within the closed funnel, the largest deterioration was before cart; checkout-to-purchase improved.

3. Funnel completeness is uncertain. Only 2,778 of 4,848 purchase sessions satisfy the full ordered journey. The other 2,070 purchase sessions, or 42.70%, must not be described as failed purchases. They contain a purchase event but do not meet the selected sequence.

4. Acquisition evidence is insufficient for budget optimization. Conflicting earliest source observations affect 64.04% of sessions. The usable acquisition comparison describes first-user source, which is a different attribution scope. Spend, reliable attributed revenue and margin are absent.

5. Matched cohorts show limited later activity, but these are calendar-week return measures. Week-1 return is 4.43% and week-4 return is 1.18%. These numbers do not mean that 98.82% of customers permanently churned.

Recommended management decision: approve a measurement diagnostic and a narrowly defined product-page experiment. Treat the January decline as a cross-device investigation. Keep acquisition budget changes and category recommendations pending stronger evidence. The recommendations below are proposed actions, not work already performed on a live store. Internal counts and derived rates come from the verified exports and report model. [D1-D4]

1 Scope And How To Read The Metrics

The analysis uses 92 daily shards from the public GA4 ecommerce sample, covering 4,295,584 event rows, 270,154 observed user identifiers and 360,129 observed sessions. Local aggregates reconcile across month/device and first-user acquisition tables. Independent checks show no funnel monotonicity violations or shard-date mismatches. These checks establish consistency of the transformations; they cannot remove inconsistencies introduced in the underlying sample. [D1-D3]

This is an analysis of exported events. An observed user identifier is not necessarily a person or a lifetime customer. The session key combines a valid user identifier and a positive GA session ID. A session is counted once even when it contains repeated events. The export does not reconstruct sessions from a timeout rule.

The principal outcome measures deliberately differ:

Metric	Count or rate
All observed sessions	360,129
Sessions containing a purchase event	4,848
Purchase-session conversion	1.346%, rounded to 1.35%
Raw purchase-event rows	5,692
Valid unique transaction IDs	4,451
Accepted transactions	4,436
Strict ordered funnel purchase sessions	2,778
Closed funnel conversion	3.607%, rounded to 3.61%

Purchase-session conversion divides purchase sessions by all observed sessions. Closed funnel conversion divides ordered funnel completions by product-view sessions. The 1.35% and 3.61% rates answer different questions and should never be compared as if one were an improved version of the other.

The funnel requires view_item, then add_to_cart, then begin_checkout, then purchase, within the same session and with strictly increasing timestamps. An event can occur between those stages; adjacency is not required. Missing stages are not inferred. This is a session-based custom funnel, whereas Google's GA4 exploration documentation describes user-based funnel counting. Its conceptual explanation of closed funnels is relevant, but its interface totals are not interchangeable with this report's counts. [D4, R2]

Monthly session metrics use the session's first observed date. Accepted transactions use their representative purchase-event date. Cross-date sessions exist, so purchase sessions and monthly transaction counts need not move identically. The local report supports monthly device comparisons and full-window first-user acquisition. It does not contain the joint month/device/source detail needed to identify which channel caused the January change.

2 The Main Funnel Opportunity Is Before Cart

Ordered step	Entry	Next	Conversion	No next step
Product view to cart	77,020	14,380	18.67%	62,640
Cart to checkout	14,380	5,187	36.07%	9,193
Checkout to purchase	5,187	2,778	53.56%	2,409

All 77,020 product-view sessions are the entry population. Cart, checkout and purchase retain 18.67%, 6.73% and 3.61% of that entry population respectively. In the dashboard, funnel widths represent these shares. The percentages beside it are step-to-step rates: for example, 53.56% is purchase among qualifying checkout sessions, not purchase among all product viewers. [D1-D2]

The 81.33% view-to-cart drop is large enough to justify investigation. However, it combines many possible behaviors: browsing without buying, comparing products, selecting unavailable variants, leaving to consider a purchase, or missing/incorrectly sequenced events. The aggregate data cannot identify which explanation dominates. It also cannot establish whether the cart button was visible or usable.

Investigate product-page behavior before selecting a solution. Review representative desktop and mobile journeys, stock and variant availability, price and delivery information, and product-selection errors. In a production implementation, capture successful cart additions separately from button clicks or errors. A higher click count alone would not show that more shoppers created valid carts.

The cart-to-checkout step also deserves attention: 63.93% of qualifying cart sessions do not reach an ordered checkout event. Across the full ordered cart-to-purchase path, only 19.32% complete, giving 80.68% observed non-completion. This is a closed, within-session measure; it excludes later-session recovery and buyers who bypass earlier tracked stages. It is not a count of permanently lost customers.

Checkout non-completion is 46.44%, or 2,409 qualifying sessions. Payment and form problems are plausible diagnostics, but this dataset has no session-linked payment-error, delivery-cost or checkout-field evidence supporting a specific cause. Product-page and cart issues should therefore be tested without claiming that checkout has been proven healthy or broken.

Only 21.39% of all sessions contain a product-view event. The remaining 78.61% do not have an observed funnel entry. This could reflect landing-page behavior, alternative routes or measurement coverage. It is not a bounce rate. Improving discovery may eventually be valuable, but a dependable product-list-to-detail analysis is unavailable here.

3 How Much Could An Improvement Matter

The following scenarios estimate the mechanical effect of improving one stage by two percentage points while holding entry volume and all other stage rates constant. They use the full historical window and produce expected additional strict funnel purchase sessions, not confirmed orders, revenue or a causal forecast. [D1-D2; calculations in D5]

Stage improved	Baseline	Assumed rate	Added funnel purchase sessions
Product view to cart	18.67%	20.67%	Approximately 298
Cart to checkout	36.07%	38.07%	Approximately 154
Checkout to purchase	53.56%	55.56%	Approximately 104

For the first scenario, 77,020 multiplied by 0.02 gives approximately 1,540 extra cart sessions. Applying the observed cart-to-purchase rate of 2,778 / 14,380 yields approximately 298 extra funnel purchase sessions. The corresponding strict funnel completion increase is about 10.71% relative to 2,778.

Two percentage points is an absolute increase. Moving from 18.67% to 20.67% is roughly a 10.71% relative increase, not a 2% relative increase. Equal percentage-point changes across stages are useful for arithmetic comparison, but they may differ greatly in cost and feasibility. These scenarios should not be added together as independent gains because simultaneous stage improvements compound.

The calculation favors investigating the earliest measured stage because of its larger eligible population. It does not prove that this intervention has the highest return on investment. Newly encouraged cart users might have lower purchase intent, and downstream conversion could change. A treatment could also create

low-quality carts or shift purchases that would have occurred later.

A commercial business case requires validated incremental accepted orders, net revenue, refunds, discounts, contribution margin and implementation cost. The current analysis cannot calculate average order value, customer acquisition cost, return on ad spend, lifetime value or profit uplift reliably. Do not attach a currency value to these scenarios by inventing an average basket size.

4 January Declined Mainly Through Conversion

Month	Sessions	Purchase sessions	Purchase conversion	Accepted transactions
Nov 2020	108,401	1,618	1.493%	1,249
Dec 2020	133,351	2,115	1.586%	2,299
Jan 2021	118,377	1,115	0.942%	888

December had the highest traffic and purchase-session conversion. January had 14,974 fewer sessions and 1,000 fewer purchase sessions than December. Traffic declined 11.23%, purchase-session volume declined 47.28%, and the conversion rate declined 40.61% relative, or 0.644 percentage points. [D1-D2]

An exact volume-first arithmetic decomposition makes the difference clearer. Applying December's purchase-session rate to January's traffic yields approximately 1,878 purchase sessions. The lower traffic explains about 237 of the 1,000-session decline; the lower rate at January's traffic explains about 763. This is a descriptive decomposition, not a causal attribution. Changing the order of the decomposition changes the assigned interaction, so the 23.7% / 76.3% split is specific to this stated method. It does not represent proven recoverable sales.

Stage rates reveal where to investigate:

Month	View to cart	Cart to checkout	Checkout to purchase	Purchases in funnel*
Nov 2020	8.09%	40.05%	50.42%	26.20%
Dec 2020	28.80%	35.88%	53.37%	72.96%
Jan 2021	18.28%	34.45%	55.74%	72.74%

*Strict funnel purchase sessions divided by all purchase sessions.

From December to January, view-to-cart declined by 10.53 percentage points, cart-to-checkout declined by 1.43 points, and checkout-to-purchase increased by 2.37 points. Product-view coverage also fell from 20.98% to 19.52% of all sessions. The observed January deterioration is therefore concentrated earlier in the measured journey, rather than showing a lower final checkout success rate among qualifying checkout entrants.

November's unusually low view-to-cart rate and funnel purchase coverage require a measurement review. Funnel coverage increased from 26.20% to about 73% in December and January. This could reflect tracking, routes or obfuscation as well as behavior. It weakens claims that one stable customer journey explains the entire three-month aggregate.

A production investigation should compare daily event coverage, release dates, tag changes, stockouts, product assortment, promotions and traffic intent. December holiday demand and January seasonality are plausible hypotheses, not established explanations. Three historical months are insufficient to identify a recurring seasonal pattern. The reduced local export also prevents a reliable source-specific explanation of the monthly decline.

5 Devices Do Not Show A Broad Mobile Failure

Device	Sessions	Traffic share	Purchase sessions	Purchase rate	Closed funnel rate
Desktop	208,942	58.02%	2,749	1.316%	3.50%
Mobile	143,185	39.76%	1,995	1.393%	3.77%
Tablet	8,002	2.22%	104	1.300%	3.59%

Mobile's purchase-session rate is 0.078 percentage points above desktop, a 5.90% relative difference. Mobile also has slightly higher measured view-to-cart and checkout-to-purchase rates. These descriptive differences do not demonstrate that the device itself causes a higher conversion rate. Visitors differ in intent, identity continuity, product selection and other unobserved characteristics. [D1-D2]

The existing session-level Wilson intervals are approximately 1.268%-1.365% for desktop, 1.334%-1.455% for mobile and 1.074%-1.572% for tablet. They describe binomial count uncertainty under an independence assumption. Repeat sessions from the same users, measurement defects and obfuscation are not captured in those intervals. Overlapping intervals are not themselves a formal test of a difference.

Both major devices deteriorated in January. Desktop purchase-session conversion declined from 1.577% in December to 0.899% in January; mobile declined from 1.607% to 0.996%. Traffic mix was almost unchanged: desktop remained about 58% and mobile about 40% in every month. A large shift toward a lower-converting device is therefore an unlikely explanation for the aggregate January decline in these records.

Prioritize desktop and mobile together because they represent 97.78% of sessions. Ensure that both work well in any experiment and examine device interactions as secondary evidence. Tablet has only 104 purchase sessions in the full window, so detailed tablet conclusions are less stable. The evidence supports cross-device diagnostics rather than a mobile-only redesign.

6 Acquisition Comparisons Are Descriptive

First-user source and medium	Sessions	Purchases	Purchase rate
Google organic	112,678	1,250	1.11%
Direct none	83,459	1,079	1.29%
Unknown source and unknown medium	74,686	1,213	1.62%
Unknown source referral	34,675	471	1.36%
Store-domain referral	28,849	585	2.03%
Google cpc	15,617	153	0.98%
Unknown source organic	10,163	97	0.95%

Two further one-session rows have zero purchase sessions and are omitted from this display; all source rows remain in the data. The table groups sessions by exported first-user acquisition. It does not show the channel that initiated each subsequent session or the incremental effect of advertising. Google distinguishes user, session and event scopes explicitly. [D2, R3]

Google organic contributes 31.29% of sessions and 1,250 purchase sessions, making it the largest named source/medium group by session volume. Google cpc has a lower observed conversion rate, but cutting paid spend on this evidence would be premature. There is no spend, validated campaign attribution, acquisition-date cohort comparison, profit measure or experiment showing incremental paid-channel value.

Store-domain referral has the highest rate among the sizable displayed groups. It may reflect high intent or site/domain routing, rather than an external acquisition channel that can be scaled. Investigate domain configuration and the meaning of this first-user label before calling it the best marketing channel. The exported data does not establish that the referral is an error.

Missingness is commercially important. Rows with an unknown first source account for 119,525 sessions, or 33.19% of traffic, and 1,781 purchase sessions, or 36.74% of purchase sessions. Unknown source is not a usable marketing channel. Separately, 230,610 sessions have conflicting source/medium observations at the earliest relevant timestamp, affecting the withheld session-source proxy. These two problems concern different fields and should not be combined or double-counted.

For a live business, repair tagging and routing, preserve unknown categories, join validated spend and orders at compatible scopes, and assess channel performance using both commercial outcomes and incrementality. Avoid optimizing to conversion rate alone; a low-volume, high-rate group can contribute fewer purchases than a larger, lower-rate group.

7 Return Activity Is Limited And Purchase Follow Up Is Sparse

Use the matched population of users first observed in complete entry weeks from 2 November to 28 December 2020. It includes 183,308 identifiers with complete week-1, week-2 and week-4 follow-up. Holding this population constant prevents the three horizons from being compared across different eligible cohort mixes. [D1-D2]

Target week	Returning users	Return rate	Repeat purchasers	Repeat purchase rate
Week 1	8,114	4.43%	37	1.79%
Week 2	3,648	1.99%	23	1.11%
Week 4	2,164	1.18%	11	0.53%

Repeat-purchase rates use a separate denominator: the same 2,064 accepted purchasers in their cohort entry week. They do not divide by all 183,308 cohort users. The small repeat-purchaser counts make forecasts and granular segment comparisons particularly unreliable.

A target-week return means activity in that particular calendar week. It is not the share who ever returned by then. A user active in week 1 and week 4 can appear in both rows, so the rows must not be summed to estimate unique repeat visitors. A user absent in week 4 may return later. Weekly offsets also differ from exact 7-day or 28-day elapsed retention.

The first observed event in this window is not necessarily a user's first-ever visit. Prior history is unavailable, making the entry cohort left-censored in that broader sense. Incomplete follow-up is excluded; unavailable rates remain blank. These definitions prevent recent cohorts from appearing to have worse retention merely because their observation time is shorter.

For the business, the pattern suggests investigating reasons to revisit and repeat purchase. Suitable hypotheses include post-purchase product recommendations and useful replenishment or new-product messages where the assortment supports them. The sample does not show consent, customer contactability, delivery completion, product purchase cycles or message exposure. It cannot justify sending campaigns or forecasting their effect. A live retention test should measure incremental accepted purchases over a mature window, alongside unsubscribes and other relevant customer guardrails.

8 Measurement Quality Limits Commercial Conclusions

Purchase rows reconcile as follows: 5,692 raw purchase events equal 4,451 distinct valid transaction IDs, plus 335 excess rows repeating valid transaction IDs, plus 906 purchase rows with invalid transaction IDs. Of the 4,451 IDs, 15 have ambiguous linkage and are excluded, leaving 4,436 accepted transactions. Invalid-ID purchase rows are 15.92% of raw purchase rows; repeated valid-ID rows are 5.89%. These are event-row diagnostics, not percentages of rejected customers. [D2-D4]

The accepted transaction definition requires one valid user and one valid session, without invalid-identity rows for that ID. It improves identity consistency but does not turn the obfuscated sample into audited financial records. Conflicting transaction revenue IDs also number 15. Accepted count validity and trustworthy monetary values require separate checks.

The closed funnel misses 42.70% of purchase sessions. Allowing equal timestamps adds only 56 completions, taking the funnel from 2,778 to 2,834. This is a 2.02% relative increase in strict completions and closes only 2.71% of the 2,070-session coverage gap. Timestamp ties alone therefore cannot explain most of the missing funnel coverage. Equal timestamps also cannot prove actual event order.

Item analysis is less dependable. There are 130,641 item rows with missing/invalid IDs and 903 item IDs linked to multiple categories. Of 386,068 view_item events, 144,192, or 37.35%, have no items array. Only 71 view_item_list events are observed. These signals undermine complete product-discovery and item-level journey comparisons. They do not demonstrate that the website had only 71 product-list displays.

The exploratory item table contains the 20 largest eligible view counts, not the whole catalog. All 20 have category ambiguity and zero same-ID ordered cart-to-purchase pairs. Zero linked pairs cannot establish that the products never sold; identifier changes, variants, missing payloads and obfuscation can break the linkage. Withhold category winners, product revenue rankings and recommendations to discontinue products.

Google's ecommerce implementation guidance describes the relevant journey events, item arrays and transaction parameters, and recommends debugging event collection. That provides a practical specification for production validation. It does not establish why this historical sample has missing or inconsistent values. [R4]

Production acceptance checks should include known test journeys across both main devices, stable item and variant mapping, unique order identifiers, correct purchase retries and event timestamps, source/medium routing, and reconciliation against a trusted order system. Define acceptable coverage with the implementation owners; do not assume every legitimate buyer must follow the selected closed path.

9 External Research Context And Benchmark Limits

Baymard's compilation reviewed on 12 October 2026 reported an average of 70.22% across 50 studies. Its consumer research identifies browsing intent and potential frictions such as extra costs, delivery speed, trust, account creation and checkout complexity. These findings are useful for selecting usability questions. They are not observed reasons for abandonment in this sample. [R5]

Our 80.68% closed cart-to-purchase non-completion rate is numerically above that external average, but the comparison does not establish underperformance. Populations, time periods, cart definitions, follow-up windows and tracking quality differ. Our 81.33% product-view-to-cart drop is an entirely different metric and must not be compared with a cart-abandonment benchmark. The current research also postdates the 2020-2021 sample. [D1, R5]

The most useful benchmark is a recent, validated production baseline using the same unit, event definitions and observation window as the proposed test. External research can inform hypotheses about total-cost disclosure or checkout usability; controlled evidence is needed to decide whether any change improves this store's outcomes.

No generic ecommerce conversion benchmark is used to grade the 1.35% purchase-session rate. A credible comparison would require aligned geography, product category, traffic mix, device mix, season, purchase definition and attribution window. Reporting that the business is good or bad against a universal rate would overstate what the available evidence supports.

10 Recommended Action Plan

First phase Measurement and diagnosis

Proposed owners are analytics engineering and ecommerce product management. Validate events and transaction identity in production, reconcile purchase events with orders, and diagnose the November coverage anomaly. Review the January decline across both desktop and mobile, using daily rates and event coverage together. Deliver an agreed metric dictionary, a measurement defect register and a baseline that can support a test. These are role suggestions rather than named commitments.

Second phase Product page experiment

Test one change to cart-action presentation after reviewing actual journeys. A suitable hypothesis is that a clear, persistent cart action beside product-selection controls increases successful cart additions. Keep prices, discounts, inventory rules and campaigns stable where practical so the effect can be attributed to the

tested change. Randomize eligible users 50/50 and keep assignment stable across visits.

Count each randomized exposed user once for the primary metric, using their first eligible session. The outcome is a successful ordered cart event in that session. Analyze all eligible randomized exposed users in their assigned arms, including those who do not engage with the new control. This proposed unit avoids treating a repeat visitor's many sessions as independent experiments.

The session baseline of 18.67% is a planning proxy. Detecting an assumed absolute increase of two percentage points with a two-sided 5% significance level and 80% power gives approximately 6,200 users per arm under an independent-proportion approximation. Recalculate using a recent production user-level baseline, exposure losses and the final design. This number is not a guarantee of adequate power for purchase or margin guardrails. [D4-D5]

Before launch, specify the sample target, analysis window, meaningful effect and guardrail bounds. Measure accepted purchase conversion per exposed user over a mature seven-day window, as well as cart errors, checkout errors and page performance. Add contribution margin, refunds and discounts only after reliable production measurement is available. Plan enough complete business-week cycles and wait for outcome maturation; duration depends on eligible exposed users per day, not all website sessions.

Check allocation balance, exposure logging and differential telemetry loss before interpreting a lift. Microsoft's research identifies sample-ratio mismatch as a symptom of experimental data-quality problems that can reverse conclusions. Investigate a mismatch rather than explaining it away. [R6]

Roll out when the predeclared primary criterion is met, the gain is commercially worthwhile, guardrails satisfy their bounds and measurement checks pass. An increase in carts with lower purchasing quality is not a successful business result. An inconclusive result is not proof of no effect. Avoid stopping a fixed-horizon test at the first favorable interim result.

Third phase Acquisition and retention decisions

After measurement is dependable, join acquisition costs with validated commercial outcomes and test incremental channel value. Develop retention hypotheses appropriate to the product cycle and mature cohort follow-up. Category optimization should wait for stable item linkage. This order prioritizes decisions the data can support and prevents attractive dashboard rankings from becoming unsupported budget decisions.

11 How A Decision Maker Should Use The Report

Overview answers whether traffic and purchase-session conversion moved together. Read volume and rate together, and remember that accepted transactions and purchase sessions count different units.

Funnel and Segments identifies which ordered stage loses the most sessions and whether that pattern persists by month and device. Stage width represents retained sessions; the rate on the right represents the transition from the preceding stage. Counts are filter-responsive. A funnel loss means failure to meet the selected sequence in the session, not necessarily a lost sale.

Acquisition describes first-user source groups. Use it to ask measurement and traffic-quality questions, while preserving the unknown groups. It is not an ROI scorecard and should not imply that changing a month or device filter produces source detail absent from the local export.

Retention uses fully observed follow-up and a matched population for comparison. State whether a number refers to return activity or repeat purchase, and always name its denominator.

Findings and Data Quality makes uncertainties visible beside performance. An apparent improvement accompanied by poorer event coverage or increased transaction ambiguity needs investigation before it is celebrated.

For an ongoing management review, track purchase sessions and accepted transactions separately, overall and closed-funnel conversion, each step's entry and loss counts, product-view coverage, source missingness, funnel purchase coverage, transaction exceptions and mature cohort outcomes. Production targets should follow baseline validation. This sample does not establish suitable numerical business targets.

12 Evidence Strength And Unanswered Questions

Strong descriptive evidence: exported counts reconcile; view-to-cart has the largest measured loss; January conversion declined across both major devices; their traffic mix was stable; source and item quality problems are substantial.

Moderate evidence for prioritization: product-page testing has a large eligible population and favorable mechanical upside under constant downstream rates. Prioritization remains conditional on measurement, implementation cost and production relevance.

Weak or unavailable causal evidence: why people left, whether a specific button design caused the drop, why January declined, which campaign created incremental orders, whether a referral label is incorrect, and which product category deserves more budget.

Questions to resolve before major commercial decisions include: Which purchase routes legitimately bypass tracked product views? Did event collection change between November and December? Which product pages and availability states are associated with unsuccessful cart attempts? Were January visitors less purchase-ready? Are transaction values reconciled to orders and refunds? Are source labels affected by domain transitions? How does repeat purchase vary by actual product cycle? None can be answered reliably from the reduced aggregates alone.

The underlying analysis was prepared on 12 October 2026 from preserved exports. This edition was revised on 13 October 2026. It does not re-query live BigQuery data or audit a current production store.

13 Reproducibility And Source Register

Internal Evidence

[D1] Product Journey/analysis/verified_summary.json
Verified totals, funnel transitions, monthly/device metrics and matched cohort summaries.

[D2] Product Journey/data/powerbi/powerbi_tables.json
Underlying report aggregates, including first-user acquisition, event coverage, quality checks and transaction counts. Counts in this analysis were reconciled across relevant tables.

[D3] Product Journey/data/provenance.json and sql/02_pipeline.sql
Source dataset, observation window, recorded query provenance and transformation definitions. The export is a reduced aggregate capture, not a full local copy of every raw event.

[D4] Product Journey/docs/metric_dictionary.md, docs/experiment_proposal.md and powerbi/Product Journey/Product Journey.SemanticModel/model.bim
Metric definitions, proposed experiment and final report calculations. Experiment details are a proposal, not experimental results.

[D5] Product Journey Business Analysis Calculations.json
New calculations for this report: monthly rates, volume-first decomposition, stage improvement scenarios, device mix, source missingness and purchase-event reconciliation. Derived from D1 and D2 without changing source data.

External Research Reviewed 12 October 2026

[R1] Google BigQuery sample dataset for Google Analytics ecommerce web implementation
https://developers.google.com/analytics/bigquery/web-ecommerce-demo-dataset
Used for source provenance, historical dates, obfuscation and internal-consistency limits.

[R2] Google Analytics Funnel exploration
https://support.google.com/analytics/answer/9327974?hl=en
Used for the conceptual distinction between open and closed funnels. Its user-based counting is different from this custom session-based report.

[R3] Google Analytics Scopes of traffic source dimensions
https://support.google.com/analytics/answer/11080067?hl=en
Used to distinguish first-user acquisition from session and event attribution.

[R4] Google Analytics Measure ecommerce
https://developers.google.com/analytics/devguides/collection/ga4/ecommerce
Used as the production specification for ecommerce events, item payloads and purchase parameters.

[R5] Baymard Institute Cart abandonment rate statistics
https://baymard.com/lists/cart-abandonment-rate
The page reviewed on 12 October 2026 reported a 70.22% average across 50 studies. Used as contextual research and a source of usability hypotheses, not a directly comparable performance target.

[R6] Fabijan and colleagues Diagnosing Sample Ratio Mismatch in Online Controlled Experiments
ACM SIGKDD 2019 Microsoft Research
https://www.microsoft.com/en-us/research/publication/diagnosing-sample-ratio-mismatch-in-online-controlled-experiments-a-taxonomy-and-rules-of-thumb-for-practitioners/
Used for the experiment validity recommendation to investigate allocation mismatch before trusting outcomes.

Calculation Conventions

Rates are ratios of summed counts, not simple averages of segment percentages. Display rounding may make percentages appear slightly inconsistent. Percentage-point changes subtract rates; relative percentage changes divide the difference by the original rate. Scenario results are expected values rounded to whole sessions for discussion. The monthly decomposition uses December as the baseline, applies its rate to January's volume, then attributes the remaining arithmetic difference to the rate change. All scenario gains refer to strict funnel purchase sessions and should remain separate from accepted-order or revenue estimates.

14 Dashboard Reconciliation - 13 October 2026

The latest supplied Power BI artifact, saved on 12 October 2026 at 20:01 IST, was inspected as a PBIX archive. All five page definitions are present and archive integrity passes. Its seven Findings & Data Quality statements agree with the report: view-to-cart loss, January decline, incomplete closed-funnel coverage, device comparisons, matched retention, source conflicts and transaction/item limitations.

The embedded experiment text also agrees: one persistent cart action, user randomization, the first eligible-session ordered cart outcome, an 18.67% planning proxy, about 6,200 users per arm, and mature seven-day accepted-purchase guardrails. This remains a proposal, not an experiment result.

Visual query bindings match the previously verified PBIX definitions.

Retention denominators remain distinct. The matched comparison uses the same 183,308 users: 8,114 / 183,308 = 4.43%, 3,648 / 183,308 = 1.99%, and 2,164 / 183,308 = 1.18%. Matrix totals use all eligible cohorts at each horizon: 10,490 / 248,347 = 4.22%, 4,234 / 227,649 = 1.86%, and 2,164 / 183,308 = 1.18%. These are compatible answers to different questions.

Quality-audit counts can overlap. Fifteen ambiguous transaction IDs and fifteen conflicting-revenue IDs must not be added as thirty independent transactions. Accepted transactions remain 4,436 after ambiguous IDs are excluded. A missing funnel sequence is not a failed purchase; source conflicts and unknown first-user source are separate diagnostics.

The preserved-export reconciliation was re-run on 13 October 2026. Event and session totals, segment sums, transaction exclusions, strict funnel ordering and cohort eligibility checks passed. These checks validate consistency of the captured aggregates, not every original BigQuery event.

Scope of this reconciliation: static page text, visual metric bindings and preserved aggregate evidence. The latest embedded DataModel differs at the binary level from the earlier verified artifact and was not independently queried or refreshed in Power BI Desktop in this review. Consequently, this edition does not claim that every live visual, filter combination or embedded model value has been freshly audited. No supported new causal conclusion was identified.

Artifact fingerprint (SHA-256): 6140ccaf52c0b5b5d84525110c1780592acf60057be81e39c78517f4e3e981a1